

1

Conditioning c

FIMr1p1 ensemble guidance

PR + T2M weekly summaries:
mean, spread, q10, q90, count

Observed predictors

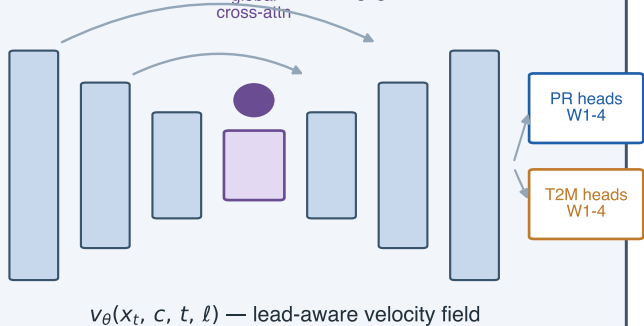
SST, SSS, soil moisture, IVT,
Z500*, U250, MJO (4 prior weeks)

Static + calendar

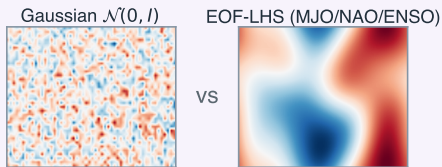
elevation, land mask, lat/lon
encodings, season, lead week

2

Conditional flow-matching generator

velocity v_{θ}

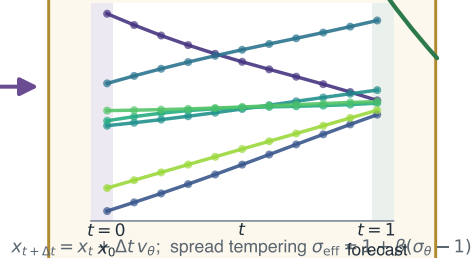
3

Physically informed stochastic prior x_0 

$$x_0 = \rho x_{\text{EOF}} + \sqrt{1 - \rho^2} x_{\text{rand}}$$

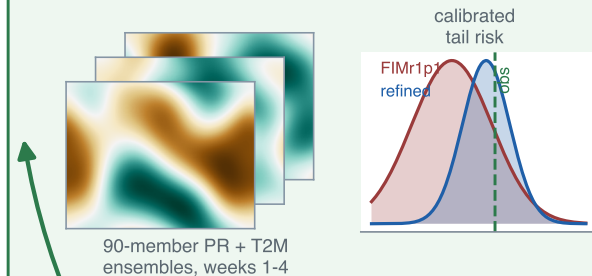
$\rho_{\text{PR}} = 0.25$, $\rho_{\text{T2M}} = 0.08$ (regime- and lead-conditioned)

ODE sampling (10 Euler steps)



4

Weekly probabilistic forecasts



Verification

CRPS	RMSE / bias / corr
BSS + calibrated BSS	spread-skill
season x lead	event tails (q95/q99)